1. **Aim/Overview of the practical:**
   To understand and implement various logical functions in Excel including:
   IF Statement
   Nested IF
   AND, OR, NOT
   IFERROR
   SUMIF, AVERAGEIF
   COUNTIF
   AVERAGEIFS

2. **Objective of the practical:**

   To understand and apply various logical and conditional functions in Excel for effective data analysis and decision-making.

3. **Steps Of experiment/practical:**

I	J	K
Total Marks	**Marks Obtained**	**Percentage**
500	455	91%
500	410	82%
500	385	77%

1. Percentage Calculation

- Cell: K2
- Formula: =J2/I2
- Purpose: Calculates percentage = Marks Obtained / Total Marks.

---

2. Grade using Nested IF

L
Grade
A+
A
B+

- Cell: L2
- Formula:

=IF(K2>=0.9, "A+", IF(K2>=0.8, "A", IF(K2>=0.7, "B+", "B")))

- Purpose: Assigns grade based on percentage.

3. Result using IF

- Cell: M2
- Formula: =IF(K2>=0.6, "Pass", "Fail")
- Purpose: Returns "Pass" if percentage ≥ 60%, else "Fail".

M
Result
Pass
Pass

---

4. Scholarship using AND in IF

- Cell: N2
- Formula: =IF(AND(K2>=0.85, H2>=0.85), "Eligible", "Not Eligible")
- Purpose: Only students with both high marks and attendance get a scholarship.

N
Schlorship
Eligible
Eligible

---

5. IFERROR Use

- Cell: O2
- Formula: =IFERROR(J2/I2, "Error")
- Purpose: Prevents error if I2 = 0 (e.g., division by 0).

O
Error Check
0.91
0.82

---

6. Attendance Check using IF

- Cell: P2
- Formula: =IF(H2>=0.8, "OK", "Low Attendance")
- Purpose: Checks if attendance is satisfactory.
-

P
Attendence check
OK
OK

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7. SUMIF for Attendance ≥ 80%

- Cell: Q2
- Formula:

=SUMIF(H:H, ">=0.8", I:I)

- Purpose: Total marks of students whose attendance is ≥ 80%.

Q
Total Marks (Attendance ≥ 80%)
2707
2252

---

8. AVERAGEIF (Marks ≥ 400)

R
Average Marks ≥ 400
453
452.3333333

- Cell: R2
- Formula:

=AVERAGEIF(J:J, ">=400", J:J)

- Purpose: Average marks of students scoring ≥ 400.

---

9. COUNTIF (Marks ≥ 400)

S
Count Students with Marks ≥ 400
4
3

- Cell: S2
- Formula:

=COUNTIF(J:J, ">=400")

- Purpose: Count of students scoring 400 or above.

---

10. IF + OR (High Performance Tag)

T
Marks Obtained ≥ 400 OR Attendance ≥ 90%
High Performance
High Performance

- Cell: T2
- Formula:

=IF(OR(J2>=400, H2>=0.9), "High Performance", "Normal")

- Purpose: If either Marks ≥ 400 or Attendance ≥ 90%, tag as High Performance.

5. Output:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
UID	Name	Computer	DBMS	Networks	AI	Java	Attendance (%)	Total Marks	Marks Obtained	Percentage	Grade	Result	Schlorship	Error Check	Attendence check
101	Aaryaman	95	88	92	85	95	95%	500	455	91%	A+	Pass	Eligible	0.91	OK
102	Paras	70	85	82	87	86	88%	500	410	82%	A	Pass	Eligible	0.82	OK
103	Prince	80	72	75	78	80	25%	500	385	77%	B+	Pass	Not Eligible	0.77	Low Attendance
104	Varun	80	58	65	57	60	85%	500	320	64%	B	Pass	Not Eligible	0.64	OK
105	Nikhil	85	95	98	99	100	48%	500	477	95%	A+	Pass	Not Eligible	0.954	Low Attendance
106	Dimpal	80	70	72	68	65	80%	500	355	71%	B+	Pass	Not Eligible	0.71	OK
107	Ankush	75	85	82	70	75	90%	500	387	77%	B+	Pass	Not Eligible	0.774	OK
108	Tushar	55	60	58	52	55	75%	500	280	56%	C	Pass	Not Eligible	0.56	OK
109	Niharika	62	65	60	58	65	85%	500	310	62%	B	Pass	Not Eligible	0.62	OK
110	Amisha	95	90	92	96	97	92%	0	470	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	Error in Data	OK

Q	R	S	T
Total Marks (Attendance ≥ 80%)	Average Marks ≥ 400	Count Students with Marks ≥ 400	Marks Obtained ≥ 400 OR Attendance ≥ 90%
2707	453	4	High Performance
2252	452.3333333	3	High Performance
1842	473.5	2	Normal
1842	473.5	2	Normal
1522	473.5	2	High Performance
1522	470	1	Normal
1167	470	1	High Performance
780	470	1	Normal
780	470	1	Normal
470	470	1	High Performance

6. Learning Outcomes (What I have learnt):

1. Learn how to use IF and Nested IF statements to evaluate conditions and return specific results.
2. Understand and apply logical operators like AND, OR, and NOT for complex condition checking.
3. Handle errors gracefully in formulas using IFERROR.
4. Use SUMIF, AVERAGEIF, and COUNTIF to perform conditional calculations on data.
5. Apply AVERAGEIFS for multi-condition average calculations to enhance data insights.